From Absolute Isolated-Zero Finiteness to Uniform Bounds on Regular Fibers

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Abstract

We give a detailed conditional proof of the projection-counting principle used in Section 11 of OpenAI's manuscript *Uniform bounds for planar polynomial limit cycles*. The hypotheses require a boundary-complete finite list of domain margins and finiteness of the isolated solutions of every fixed system in a specified differential and algebraic closure. Under these hypotheses, a regular parameterized equation system has a uniform bound on the number of connected components of its fibers. The proof uses a proper function, generic linear perturbations, and induction on the parameter dimension. A single perturbation is chosen for a countable collection of witness fibers, avoiding both Fubini's theorem and a separate argument at exceptional parameters. We spell out the closure inheritance and the Jacobian calculation. Two counterexamples show why neither finiteness for the original total zero set nor finiteness for the full closure without boundary completeness suffices. This note audits one conditional implication; it does not establish the analytic hypotheses for polynomial-flow matching systems or certify the full Hilbert-sixteenth manuscript.

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1 Scope, provenance, and the logical target

The target is the implication

$$\text{absolute isolated-zero finiteness for a full finite closure}$$
$$\Downarrow$$
$$\text{uniform component bounds for regular fibers.}$$

It is Theorem 11.2 in the source manuscript [1], with its hypotheses in Definition 11.1. We work from the public repository at the fixed revision

`4eaee87548250b2682affda7d05d947c05554bae`.

The relevant source is `build/sections/11-counting.tex`, lines 22–217 at this revision. Pinning a revision records what is being discussed; it is not a statement that other parts of that revision have been checked.

The source uses a generic tilt for almost every parameter slice and then extends the bound to exceptional parameters. The proof below has a different organization. If the desired bound fails, choose a *countable* sequence of witness fibers. Sard's theorem allows one tilt that is good on every member of this sequence. This is enough for the contradiction and removes the exceptional-parameter step.

All mathematical arguments in this note are supplied for examination. This document is AI-assisted, has not received a recorded independent human referee report, and is not a formal proof-assistant certificate. It makes no claim that the full Hilbert-sixteenth proof has been verified. In particular, the source's Theorem 11.4, which asserts that the required matching systems satisfy the closure hypothesis, is *not* an input proved here.

We use the following standard background facts:

(i) the implicit function theorem and the regular-level-set theorem;

(ii) Sard's theorem for smooth maps between finite-dimensional second-countable manifolds;

(iii) the classification of connected one-dimensional manifolds without boundary as circles or open intervals, up to diffeomorphism.

These are background tools, not additional results about polynomial flows. The final counterexample also uses the standard fact that a real algebraic set has finitely many connected components. Sard's original paper is listed in the references; conventional background sources are listed for the manifold and semialgebraic facts [2, 3, 4]. No particular textbook theorem number is used.

2 The hypotheses, with domains retained

Write the coordinates as

$$(p, x) \in \mathbb{R}^a \times \mathbb{R}^n.$$

The letters p and x indicate their roles in a projection. They do not indicate different kinds of analytic variables. In an absolute zero test, the remaining parameter coordinates are also unknowns.

Definition 2.1 (Open data and boundary-complete margins)**.** An open analytic datum consists of an open set $U \subset \mathbb{R}^a \times \mathbb{R}^n$, a real analytic map $F : U \to \mathbb{R}^m$, and a finite list of real analytic functions

$$g_1, \dots, g_s : U \longrightarrow (0, \infty).$$

The list is *boundary-complete*, or satisfies *positive-margin completeness*, if the following holds: whenever $z_\nu \in U$ converges to a finite point z_* and

$$\inf_\nu g_j(z_\nu) > 0 \qquad (1 \le j \le s),$$

then $z_* \in U$. The equations are $F = 0$; the margins record its original open restrictions, including coordinate-chart boundaries.

Boundary completeness means precisely this sequential condition. No compactness of U, nor extension of F across a vanishing margin, is assumed. A restriction $h \neq 0$ is recorded by the positive margin h^2. Empty zero sets and empty domains are allowed.

Definition 2.2 (Finite generated closure)**.** Starting from the coordinate functions, F_i, g_j, and fixed real constants, permit the following finite operations:

(a) introduce finitely many additional independent real coordinates, pulling back the old functions to the resulting product domain;

(b) form finite sums and products, and take finitely many partial derivatives in independent coordinates;

(c) invert a generated function h on a domain where $h \neq 0$, retaining that restriction as a margin;

(d) append finitely many equations and positive margins made from generated functions, while retaining every earlier equation and open restriction.

Coordinates may be regrouped as parameters and unknowns. Fixed real values may be assigned to coordinates; equivalently, one retains those coordinates and appends equations $z_i = c_i$. Clearing a denominator is permitted only while retaining the corresponding nonzero condition.

A *descendant system* is any system obtained by finitely many such operations, possibly repeated. Adjoining an arbitrary new analytic function is not an allowed operation. Any analytic graph ties needed in an application must already be among the data or be obtained by the stated operations.

Every particular descendant has finitely many coordinates, equations, and margins. There is no common upper bound on their numbers or on the derivative orders as the descendant varies. Fixed real constants are unrestricted: the generic tilts selected below are allowed constants.

To avoid ambiguity about specialization, a partial derivative is taken in an ambient independent coordinate before imposing an equation fixing that coordinate. Retaining the fixing equation gives the same solution set as eliminating the fixed coordinate, and preserves isolation.

Definition 2.3 (Absolute isolated-zero property for the closure)**.** The datum has property AZ if, for every fixed descendant system, its solution set has finitely many isolated points in its specified open domain when *all remaining coordinates* are treated as unknowns. An isolated solution is isolated in the total solution set, including all auxiliary coordinates and all retained equations.

The quantifiers in this property are

$$\forall \text{ fixed finite descendants } E, \qquad \#\{\text{isolated total solutions of } E\} < \infty.$$

There is no number required to work simultaneously for all E. The theorem will infer a uniform bound over p for *one fixed original system*. These are different assertions.

Lemma 2.4 (Inheritance of the hypotheses)**.** *Boundary completeness is preserved by introducing independent coordinates, by a finite succession of generated strict restrictions with their margins retained, and by taking fixed-coordinate slices. Property* AZ *is inherited by every descendant, with respect to its own finite generated closure.*

Proof. For additional coordinates, a convergent sequence projects to a convergent sequence in the old domain. The original margins put the limit in that domain, and the new coordinates have their given finite limits. For a new condition $h > 0$, a positive lower bound on its margin implies $h(z_*) > 0$ by continuity after the old domain has been recovered. For $h \neq 0$, use the margin h^2. Each new function is analytic on the domain already constructed, so this argument applies one stage at a time, including inverses.

For a slice obtained by fixing coordinates, append their fixed values to the convergent sequence of the remaining coordinates. Apply the old completeness condition and then take the slice again.

Finally, a finite sequence of permitted operations on a descendant is still a finite sequence of permitted operations on the original datum. All original equations and restrictions remain present. Thus the absolute zero assertion for the original closure includes every system in the descendant's closure. □

3 The conditional uniformity theorem

Theorem 3.1 (Uniform regular-fiber bound)**.** *Let (U, F, g) satisfy Definitions 2.1–2.3, with $0 \leq m \leq n$. Suppose that*

$$\operatorname{rank} F_x(p, x) = m \quad \text{at every } (p, x) \in U \text{ with } F(p, x) = 0.$$

Then there is a finite nonnegative integer N, depending on this fixed datum, such that every fiber

$$M_p = \{x : (p, x) \in U,\ F(p, x) = 0\}$$

has at most N connected components. If $m = n$, every fiber has at most N points.

For $m = n$, the regular-level-set theorem makes each fiber discrete. For $m < n$, the fibers are smooth manifolds of dimension $n - m$; they need not be compact or connected. The theorem gives no effective formula for N. It makes no assertion about singular fibers in which the displayed rank condition fails.

One can apply the theorem separately on the finitely many charts specified by a nonzero $m \times m$ minor of F_x. The square of the chosen minor is an additional margin. A connected component of one chart's fiber belongs to only one component of the original fiber, so summing the bounds for a finite covering bounds the original components. Such a reduction is useful in applications; it does not permit an infinite collection of charts.

The proof needs two mechanisms. A proper function gives a critical point on every component. A common generic linear tilt makes those critical points regular solutions of a finite multiplier system. In the square case, allowing one parameter to vary first reduces the number of parameters in that multiplier system.

4 Proper functions and regular critical-point systems

4.1 A proper function from the margins

Lemma 4.1 (Proper barrier)**.** *For boundary-complete open data define*

$$\rho(p,x) = 1 + |p|^2 + |x|^2 + \sum_{j=1}^{s} g_j(p,x)^{-2}. \tag{1}$$

On the equation locus, and on any fixed-coordinate slice of that locus, ρ has compact sublevel sets. If z denotes the varying coordinates of a slice, then for every fixed vector ℓ the function $\rho + \ell \cdot z$ also has compact sublevel sets and is bounded below. Its restriction to every nonempty connected component of that slice attains a minimum.

Proof. A sublevel $\rho \le C$ bounds all coordinates and gives $g_j \ge C^{-1/2}$ whenever the sublevel is nonempty. A sequence in that sublevel has a convergent subsequence. Boundary completeness puts its limit back in U, and continuity preserves $F = 0$, the fixed coordinates, and the sublevel inequality. The sublevel is therefore compact in the ambient Euclidean space.

For a linear tilt, use

$$|z|^2 + \ell \cdot z \ge \frac{1}{2}|z|^2 - \frac{1}{2}|\ell|^2.$$

All inverse-margin terms remain nonnegative. A tilted sublevel thus still bounds the varying coordinates and keeps every margin bounded away from zero. The fixed coordinates are constants, so the same compactness argument applies. The inequality also gives a lower bound.

Let C_0 be a nonempty connected component of a slice and choose $z_0 \in C_0$. Components are closed in that slice. Intersecting C_0 with the compact sublevel at the value of z_0 gives a nonempty compact set on which the function attains a minimum. This minimum is also the minimum on C_0. □

All functions used in (1) belong to the generated closure. At a later induction stage, the barrier is rebuilt using the squares of *all* coordinates then present, including multipliers, and the inverse squares of all retained margins. Reusing a barrier that omitted new unbounded coordinates would not establish the needed properness.

4.2 The bordered Jacobian

Let $W \subset \mathbb{R}^q$ be open, let $G : W \to \mathbb{R}^r$ be smooth, and suppose DG has row rank r on $M = \{G = 0\}$. Let $h : W \to \mathbb{R}$ be smooth. At a constrained critical point of $h + \ell \cdot z$, there is a unique $\lambda \in \mathbb{R}^r$ such that

$$\nabla h(z) + \ell = DG(z)^T \lambda.$$

Existence follows because the gradient annihilates the tangent space $\ker DG(z)$; uniqueness follows from the injectivity of $DG(z)^T$.

Define

$$K_\ell(z,\lambda) = \begin{pmatrix} G(z) \\ \nabla h(z) + \ell - DG(z)^T \lambda \end{pmatrix}. \tag{2}$$

There are $r + q$ equations in $q + r$ unknowns. At a zero set

$$A = DG(z), \qquad H = D_z^2 \left(h + \ell \cdot z - \sum_{i=1}^{r} \lambda_i G_i \right).$$

Direct differentiation gives the square matrix

$$D_{(z,\lambda)}K_\ell = \begin{pmatrix} A & 0 \\ H & -A^T \end{pmatrix}. \tag{3}$$

Lemma 4.2 (Regular multipliers and Morse critical points)**.** *The matrix (3) is invertible if and only if the Hessian of $(h + \ell \cdot z)|_M$ at the critical point is nondegenerate.*

Proof. The constrained Hessian is the restriction of the symmetric bilinear form H to $T_z M = \ker A$. This follows by differentiating the constraint along a local parametrization of M: the second derivatives of the parametrization in the Hessian of $h + \ell \cdot z$ are cancelled by the corresponding terms in $\sum_i \lambda_i D^2 G_i$.

The kernel equations for (3) are

$$Av = 0, \qquad Hv = A^T \eta.$$

If the restricted form is nondegenerate, testing the second equation against every $w \in \ker A$ gives $v = 0$. Then $A^T \eta = 0$, so $\eta = 0$ because A has full row rank.

Conversely, if the restricted form is degenerate, there is a nonzero $v \in \ker A$ with $w^T H v = 0$ for every $w \in \ker A$. Hence

$$Hv \in (\ker A)^\perp = \operatorname{im} A^T.$$

For some η, (v, η) is a nonzero kernel vector of (3). This proves the equivalence. □

4.3 A common generic tilt for countably many manifolds

Here a function is *Morse* when all its critical points are nondegenerate. Distinct critical points need not have distinct values.

Lemma 4.3 (Generic linear tilt)**.** *In the preceding setting, for almost every $\ell \in \mathbb{R}^q$, the function $(h + \ell \cdot z)|_M$ is Morse. Consequently, for a countable collection of such pairs (M_k, h_k) in the same ambient dimension q, a single ℓ can be chosen so that every $(h_k + \ell \cdot z)|_{M_k}$ is Morse. No bound on the number of components of M_k is required.*

Proof. The manifold $M \times \mathbb{R}^r$ has dimension $(q - r) + r = q$. Consider the smooth map

$$L : M \times \mathbb{R}^r \longrightarrow \mathbb{R}^q, \qquad L(z, \lambda) = DG(z)^T \lambda - \nabla h(z).$$

Its fiber over ℓ is precisely the critical-point system (2). At such a point, for a tangent vector (v, η) with $Av = 0$, its derivative is

$$DL(v, \eta) = -Hv + A^T \eta.$$

Since source and target both have dimension q, this derivative is surjective exactly when its kernel is zero. By Lemma 4.2, that is exactly the nondegeneracy of the corresponding constrained critical point.

Sard's theorem says that the critical values of L have Lebesgue measure zero. Thus every regular value ℓ makes all constrained critical points nondegenerate. A value not in the image has no critical points and satisfies the assertion vacuously.

The manifold is second countable: M is an embedded submanifold of an open Euclidean set. Sard can therefore be applied through a countable atlas even when there are infinitely many components. For a countable collection, take the complement of the union of its measure-zero bad sets. This complement has full measure, and any single ℓ in it has all the required properties. □

The lemma only concerns nondegeneracy. The existence of a minimum on every component comes separately from Lemma 4.1. Keeping these two roles distinct is essential on noncompact manifolds.

5 Square systems: induction on parameter dimension

Lemma 5.1 (Monotonic parameter on a regular solution curve)**.** *Suppose $F(b,t,x) = 0$ has n equations in $x \in \mathbb{R}^n$, and F_x is invertible on its zero set. For a fixed b, let*

$$C_b = \{(t,x) : (b,t,x) \in U,\ F(b,t,x) = 0\}.$$

Every connected component of C_b meets a slice $t = t_0$ in at most one point.

Proof. The full row rank of $F_{(t,x)}$ makes C_b a one-dimensional embedded manifold without boundary. On its tangent space,

$$F_t\, dt + F_x\, dx = 0.$$

If $dt = 0$, then $dx = 0$; thus dt is nonzero on every nonzero tangent vector.

A connected component is diffeomorphic to a circle or an open interval. The circle case is impossible, since the real-valued function t would attain a maximum and a minimum. In an interval coordinate, the derivative of t is continuous and never zero, so it has constant sign. Hence t is strictly monotone on the component, which proves the claim. □

Proposition 5.2 (Uniform point bound for square systems)**.** *Theorem 3.1 holds when $m = n$.*

Proof. We induct on the number a of parameters. The induction statement is simultaneous for all unknown dimensions n and for all data satisfying the hypotheses. Increasing the number of unknowns in an auxiliary system is therefore harmless; the quantity that decreases is a.

If $n = 0$, every fiber is empty or the unique point of $\mathbb{R}^0$, so the assertion is immediate. We may assume $n \geq 1$.

For $a = 0$, invertibility of F_x makes every solution isolated. Property AZ, applied to the original system itself, gives the required finiteness.

Assume the assertion proved for $a-1$ parameters and every unknown dimension. Write $p = (b,t)$ with $b \in \mathbb{R}^{a-1}$. If no uniform bound existed, choose a sequence

$$p_k = (b_k, t_k), \qquad \#M_{p_k} > k, \qquad k = 1, 2, \ldots. \tag{4}$$

There is no requirement that the parameters or their b-coordinates be distinct. An infinite fiber can be used repeatedly. Equivalently, one may select $k+1$ distinct points from each chosen fiber.

For each k, form the entire curve C_{b_k} of Lemma 5.1, allowing t to vary over its original domain. That lemma implies that C_{b_k} has more than k components: two of the selected points at the same t_k cannot belong to the same curve component.

Apply Lemma 4.3 to the countable collection

$$(C_{b_k},\ \rho(b_k, \cdot))\,, \qquad y = (t,x) \in \mathbb{R}^{n+1}.$$

Choose one vector $\ell \in \mathbb{R}^{n+1}$ that makes $\rho(b_k, y) + \ell \cdot y$ Morse on every C_{b_k}. This choice concerns each entire curve, even if it has infinitely many components. By Lemma 4.1, each component has a minimum. Every such minimum is nondegenerate.

With this single ℓ fixed, define

$$K(b,y,\lambda) = \begin{pmatrix} F(b,y) \\ \nabla_y \rho(b,y) + \ell - F_y(b,y)^T \lambda \end{pmatrix}, \qquad \lambda \in \mathbb{R}^n. \tag{5}$$

There are $2n+1$ equations and $2n+1$ unknowns $z = (y, \lambda)$, but only $a-1$ parameters b. Let

$$\Delta(b,z) = \det D_z K(b,z),$$

and use the open domain

$$U_K = \{(b, t, x, \lambda) : (b, t, x) \in U,\ \Delta(b, t, x, \lambda) \neq 0\}. \tag{6}$$

Retain the original margins and add Δ^2 as a margin.

We check the induction hypotheses rather than treating this as an automatic reduction.

(i) K is a finite system in the generated closure. The function ρ uses coordinates, sums, products, and inverses of the positive margins. Its gradient uses partial derivatives; the multiplier term uses additional coordinates and products. The determinant Δ is a polynomial in derivatives of these expressions. The chosen ℓ is a fixed real constant.

(ii) The new domain is boundary-complete with its retained margins. This is Lemma 2.4; in particular, a bounded sequence whose new margin Δ^2 has a positive lower bound cannot leave the regular Jacobian locus at its limit.

(iii) Property AZ is inherited for the *whole closure* of this new system. Any later differentiation, inversion, multiplier construction, or specialization is still a finite descendant of the original system.

(iv) $D_z K$ is invertible on U_K, by its definition.

The induction hypothesis therefore gives a finite bound N_K on the number of solutions of $K = 0$ in U_K, for every b.

At a minimum on C_{b_k}, the multiplier exists and is unique. Lemma 4.2 puts this solution in U_K. Distinct components give distinct y's and hence distinct (y, λ)'s. Consequently the b_k-fiber of this *one fixed* system has more than k points. Taking $k > N_K$ contradicts the induction bound. □

Remark 5.3 (No exceptional parameters are discarded). The witness sequence in (4) is chosen before the tilt. Its members may lie entirely in a lower-dimensional or measure-zero parameter set. The common-tilt lemma treats those particular slices directly. Thus the argument does not assume that large fibers occur at generic parameters.

6 Positive-dimensional fibers

Proof of Theorem 3.1. The square case is Proposition 5.2. Assume $m < n$, and suppose that no uniform component bound exists. Choose parameters p_k such that M_{p_k} has more than k connected components. Again repetitions are allowed.

Each M_{p_k} is a smooth manifold of dimension $n - m$. Lemma 4.3, with $q = n$ and $r = m$, provides a common $\ell \in \mathbb{R}^n$ such that

$$x \longmapsto \rho(p_k, x) + \ell \cdot x$$

is Morse on every witness fiber. Its minimum on each component exists by Lemma 4.1.

For all parameters p, use the same square multiplier system

$$J(p, x, \lambda) = \begin{pmatrix} F(p, x) \\ \nabla_x \rho(p, x) + \ell - F_x(p, x)^T \lambda \end{pmatrix} = 0, \qquad \lambda \in \mathbb{R}^m. \tag{7}$$

It has $n + m$ equations in $n + m$ unknowns (x, λ). Restrict to

$$\det D_{(x,\lambda)} J \neq 0$$

and retain the square of this determinant as a positive margin. Exactly the checks in the proof of Proposition 5.2 show that this is admissible open data with inherited property AZ. The square result, already proved for *every* parameter dimension a, gives a uniform point bound for this fixed system.

Every component of each witness fiber supplies a nondegenerate minimum and therefore a solution in this regular locus. Distinct components supply distinct solutions. There are more than k such solutions at p_k, a contradiction for large k. If $m = 0$, the construction has no multiplier coordinates; it becomes $\nabla_x \rho + \ell = 0$, and the same proof applies. □

The two dimension reductions have different purposes. In the square case a parameter is turned into a curve variable, reducing the parameter count by one. For a positive-dimensional fiber, the multiplier equations first make the problem square at the *same* parameter count. There is no circular invocation of the general component theorem.

7 Quantifiers and the scope of the conclusion

The following order is sufficient for the square-case contradiction:

$$\begin{aligned} &\text{assume unbounded fiber cardinalities} \\ \Longrightarrow\quad &\text{choose all witnesses } (b_k, t_k) \\ \Longrightarrow\quad &\text{choose one } \ell \text{ good for every witness curve} \\ \Longrightarrow\quad &\text{form one finite system } K_\ell \\ \Longrightarrow\quad &\text{obtain one finite } N_{K_\ell} \\ \Longrightarrow\quad &\text{choose } k > N_{K_\ell}. \end{aligned}$$

Although the chosen tilt can depend on the whole witness sequence, the system contains only the finitely many entries of that vector as constants. It does not contain the sequence as an infinite list of equations. The bound is permitted to depend on the tilt; it is still one fixed bound, contradicted by the infinitely many witnesses.

Assertion	Role
Each fixed finite descendant has finitely many isolated total solutions	Required hypothesis
A single bound for all descendants	Not required
One common tilt for countably many witness manifolds	Supplied by Sard and countable intersection
One common tilt good for every parameter value	Not required
A finite number of components before the tilt is chosen	Not required
All original open restrictions survive each construction	Required throughout
Uniform estimates over all derivative orders	Not required
An effective or computable component bound	Not supplied

At every induction step the new expressions involve only finitely many derivatives. Parameter dimension decreases in the square induction, so only finitely many such steps occur. This finiteness does not weaken the universal closure assumption: one must know that AZ applies to the particular, possibly complicated descendants created by those steps.

Corollary 7.1 (The conditional interface to a cycle-counting argument)**.** *Suppose a separate argument supplies finitely many fixed admissible matching systems with property* AZ*, and represents every relevant orbit, apart from at most T exceptions, by a point in one of finitely many regular minor charts. Suppose also that distinct represented orbits cannot belong to the same connected component of a fiber in any one chart, at a parameter value common to the orbits being counted. Then the number of those orbits is bounded by T plus the sum of the component bounds of the charts.*

Proof. Apply Theorem 3.1 separately to the finitely many charts. In one chart, different orbits occupy different components, so its component bound bounds the number of orbits represented there. Taking the union over charts can only overcount orbits. Finally add the T exceptions. □

This corollary exposes the remaining obligation in the source application. The existence of the finite matching description, regularity, separation of orbit representations into components, and property AZ for *every required finite descendant* are additional mathematical assertions. They are not consequences of Theorem 3.1. Nor does this note address the passage from hyperbolic cycles to arbitrary isolated cycles or the handling of the entire plane.

8 Two counterexamples delimiting the hypotheses

8.1 A: no isolated total zeros, but unbounded finite fibers

Let

$$U = \{(p, x) : 0 < p < 1,\ 0 < x < 1\}, \qquad F(p, x) = \sin(x/p).$$

Take p as the parameter and x as the unknown. Use the complete margin list

$$p, \quad 1 - p, \quad x, \quad 1 - x.$$

If these margins stay positively bounded below along a convergent sequence, its limit is still in U. Thus the domain condition holds. The function is real analytic there, and at every zero,

$$x = k\pi p, \qquad k \geq 1, \qquad F_x(p, x) = \frac{(-1)^k}{p} \neq 0.$$

The fiber cardinality is

$$\#M_p = \#\{k \geq 1 : k\pi p < 1\} = \left\lceil \frac{1}{\pi p} \right\rceil - 1,$$

which is finite for every $p > 0$ and unbounded as $p \downarrow 0$.

On the other hand, the total zero set has *no isolated points.* Each of its points lies on one of the open analytic curve segments

$$\{(p, k\pi p) : 0 < p < 1/(k\pi)\}.$$

Finiteness of the isolated points of this total set is therefore vacuously true.

The full closure detects the failure. Append the polynomial equation $x = 1/2$. The resulting system has the infinitely many solutions

$$(p, x) = \left(\frac{1}{2\pi k}, \frac{1}{2}\right), \qquad k = 1, 2, \ldots.$$

They are isolated. Indeed, the Jacobian determinant of $(F, x - 1/2)$ in (p, x) is

$$F_p = -\frac{x}{p^2} \cos(x/p),$$

which is nonzero at every displayed point. Thus property AZ fails for one very simple descendant.

This example preserves both the regularity and the boundary-complete domain hypotheses. It disproves replacing Definition 2.3 by finiteness of isolated points in the original total zero set, even if every original parameter fiber is separately known to be finite.

8.2 B: isolated-zero finiteness without boundary completeness

Let

$$D = \bigcup_{k\geq 1} (3^{-k},\, 2\cdot 3^{-k}), \qquad U = D \times \mathbb{R}, \qquad F(x,y) = y.$$

There are no parameters: $a = 0$, $n = 2$, and $m = 1$. Use only the constant margin $g_1 = 1$. The derivative $DF = (0, 1)$ has row rank one everywhere. Its zero set is

$$M = D \times \{0\},$$

which has infinitely many connected components.

Nevertheless, this datum has property AZ for the precise generated closure of Definition 2.2. To see this, start with the coordinates, $F = y$, and $g_1 = 1$. Every expression generated by finitely many sums, products, partial derivatives, additional coordinates, fixed real constants, and permitted inverses is a rational function of the full coordinate tuple. This follows inductively; in particular, a partial derivative of a rational function is rational.

Take any fixed finite descendant. Clear its finitely many denominators while retaining their nonzero restrictions. Its equations become finitely many polynomial equations. Its permitted domain is an open subset of the full Euclidean coordinate space: it retains the original restriction $x \in D$ and any finitely many additional strict restrictions. Fixing a coordinate can be represented by an additional polynomial equation.

If a solution is isolated after restricting that algebraic zero set to this open domain, it is already an isolated point of the algebraic zero set itself. In fact, a sufficiently small ball about the point is contained in the domain, so any arbitrarily nearby algebraic solutions would also be solutions of the restricted system. A real algebraic set has finitely many connected components, hence finitely many isolated points. It follows that every such descendant has finitely many isolated solutions. This verifies AZ, despite the infinitely many components of M.

The missing hypothesis is boundary completeness. For any fixed k, approach 3^{-k} from inside $(3^{-k}, 2\cdot 3^{-k})$, keeping $y = 0$. The sequence is bounded and the margin is always one, but its limit is outside U. The proposed barrier is

$$\rho(x,y) = 2 + x^2 + y^2.$$

On each component of M, it has infimum $2 + 3^{-2k}$ but no minimum. Lemma 4.1 therefore fails exactly where the counting argument needs it.

The restriction to the *generated* closure matters in this example. We have not claimed finiteness after adjoining arbitrary new analytic functions on U. Such an operation is absent from Definition 2.2. The example shows that even the full stated algebraic and differential closure cannot compensate for an arbitrary open domain whose boundary is not recorded by its margins.

9 Review checkpoints and source comparison

A referee can check the conditional implication in the following order:

1. Verify that the domain condition makes the inverse-margin barrier proper; inspect Lemma 4.1 and Example B.
2. Verify the square critical-point Jacobian (3), including the uniqueness of multipliers.
3. Apply Sard to the map L in Lemma 4.3; check that its source and target both have dimension q.
4. Verify that t is globally injective on every solution-curve component in Lemma 5.1; local graph existence alone would not justify that conclusion.
5. Follow the witness sequence through (5). The critical system must be a single fixed system with $a-1$ parameters, not one separately chosen system for each k.
6. Check all inherited margins and equations before using the induction hypothesis.
7. Reduce general fibers to the square system (7), at the unchanged parameter dimension.

Pinned source, Section 11	Exposition in this note
Definition 11.1; lines 22–56	Definitions 2.1–2.3
Proper function; lines 88–105	Lemma 4.1
Curve injectivity; lines 112–124	Lemma 5.1
Generic critical system; lines 126–159	Lemmas 4.2 and 4.3
Parameter induction; lines 161–185	Proposition 5.2, using countable witnesses
General fibers; lines 187–203	Section 6
Closure warning; lines 207–216	Section 7 and Example A

The original proof's almost-everywhere parameter argument and its implicit-function extension to exceptional parameters are not used here. This alternate organization does not relax the substantive closure hypothesis. A future audit of the full manuscript must establish that hypothesis for the actual flow-passage systems and their generated descendants, with their true open domains.

References

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